

Individual assignment 5

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(janitor)
library(readxl)
library(snakecase)
library(scales)

# file organization
library(here)

# arrange plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "Aquatic Insects")

# site information
```

```
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

Cleaning data

```
# creating a new object from sites
ncos_sites <- sites |>
  # filtering to only include NCOS quarterly sampling
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

# creating a new object
water_quality_clean <- water_quality |>
  # cleaning column names
  clean_names() |>
  # use a filtering join, only include sites in ncos_sites
  semi_join(ncos_sites,
            by = "site") |>
  # creating new column to join with later data frames
  unite("date_site", date, site,
        remove = TRUE) |>
  # grouping by date/site
  group_by(date_site) |>
  # calculating median, pH, salinity, and DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungrouping data frame
  ungroup() |>
  # filtering out outliers
  filter(date_site != "2022-08-12_NVBR")

# create a new object from taxon_list
taxon_list_clean <- taxon_list |>
  # change names in verbatim_name to snake case
```

```
mutate(verbatim_name = to_any_case(verbatim_name,
                                   case = "snake"))
```

```
# create a new object from aq_ins
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filter join to only include sites matching with ncos_sites
semi_join(ncos_sites,
          by = "site") |>
# rename column date_on_vial to date
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
# filter to only include FB250/FB 250 samples
filter(sample_type %in% c("FB250", "FB 250")) |>
# pivot data frame
pivot_longer(
  # columns to make longer
  cols = ostracod:annelida_polychaete,
  # new column for the old column names
  names_to = "taxon",
  # new column for the values in each of the old columns
  values_to = "abundance"
) |>
# group by date, site, and taxon
group_by(date, site, taxon) |>
# summarize to find average abundance per liter
summarize(ave_lit = sum(abundance, na.rm = TRUE) / 7.5) |>
# ungroup
ungroup() |>
# combine date and site into date_site, don't remove date and site columns
unite("date_site", date, site, remove = FALSE) |>
```

```

# left join water_quality_clean by date_site column
left_join(water_quality_clean, by = "date_site") |>
# left join taxon_list_clean by taxon(aq_ins)/verbatim_name(taxon_list) column
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# mutate to create a column for full names of site codes
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelp Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
))

```

```

# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
# filter to only include copepods
filter(taxon == "copepod") |>
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))

```

Visualizations from class

```

# base layer
salinity <- ggplot(data = aq_ins_clean,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
    y = med_sal)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
  width = 0.1,
  shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent means
stat_summary(fun = median,

```

```

        color = "red",
        size = 1) +
# relabelling x- and y-axes
labs(x = "Site",
     y = "Median salinity (ppt)",
     title = "Site salinity") +
# different theme
theme_bw()

```

```

# base layer
copepods <- ggplot(data = copepods,
                  # x-axis: site
                  # y-axis: mean abundance/L
                  mapping = aes(x = site_full,
                               y = log_ave_lit)) +
# first layer: points to represent observations
geom_jitter(height = 0,
            shape = 21,
            width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
            fun = "median",
            size = 4,
            color = "blue") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(10)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Copepods") +
# different theme
theme_bw()

```

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

In part 1d, you should end up with a three panel figure. All three panels should have site on the x-axis, but each panel will have a different y-axis.

In part 1c, you will create one component of the three panel figure. Provide a list of the axes and geometries you will need to create a figure in 1c showing differences between sites in ostracod abundance.

- x-axis: `site_full`
- y-axis: `log_ave_lit`
- geometry to represent average abundance/L: `geom_jitter`
- geometry to represent medians: `stat_summary`

b. Wrangle your data

The `aq_ins_clean` data frame will need some further wrangling before you create the figure you described in part 1a.

In this section, create the data frame you will need to make your figure.

At the end of your code chunk, display 10 random rows from your data frame using `slice_sample()`.

```
ostracods <- aq_ins_clean |>
  filter(taxon == "ostracod") |>
  mutate(log_ave_lit = log(ave_lit + 1))

slice_sample(ostracods, n = 10)
```

A tibble: 10 x 24

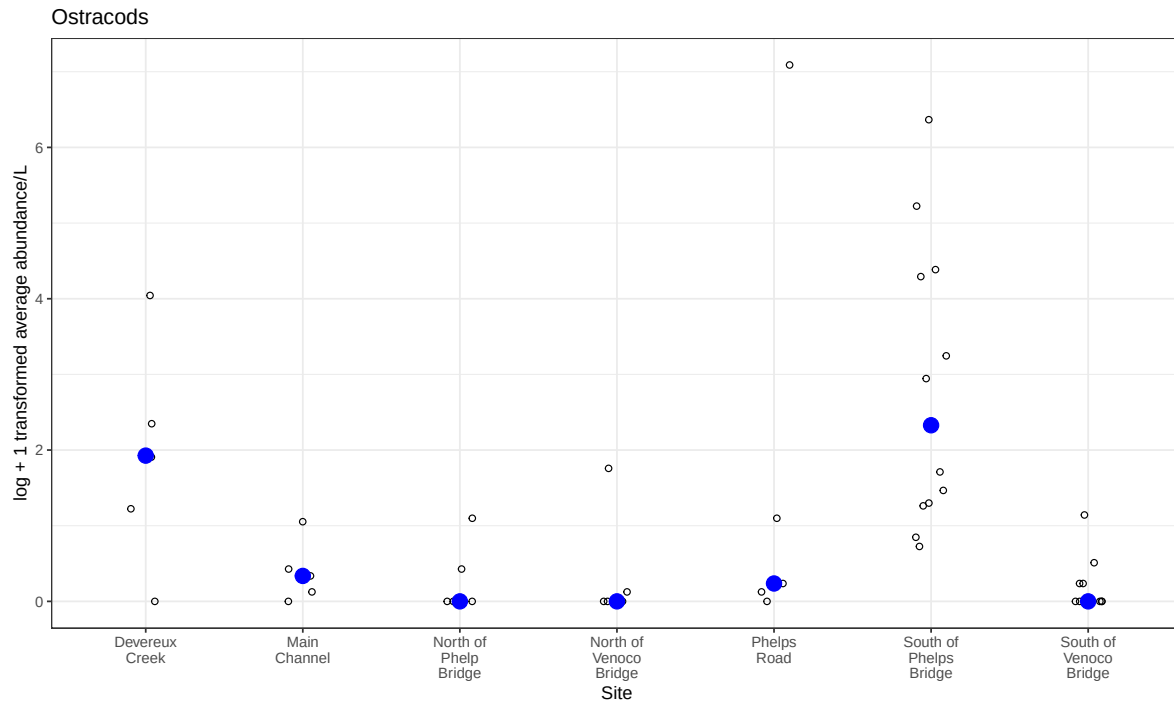
	date_site <chr>	date <dtm>	site <chr>	taxon <chr>	ave_lit <dbl>	med_ph <dbl>	med_sal <dbl>	med_do <dbl>
1	2022-01-21_NEC	2022-01-21 00:00:00	NEC	ostr~	0	NA	NA	NA
2	2023-02-12_NEC	2023-02-12 00:00:00	NEC	ostr~	2.13	NA	NA	NA
3	2020-01-21_NDC	2020-01-21 00:00:00	NDC	ostr~	2.4	6.53	0.3	7.6
4	2022-05-13_NVBR	2022-05-13 00:00:00	NVBR	ostr~	0	9.72	74.2	11.4
5	2022-05-05_NEC	2022-05-05 00:00:00	NEC	ostr~	0.267	9.45	63.8	7.3
6	2022-01-28_NDC	2022-01-28 00:00:00	NDC	ostr~	9.47	7.65	2.00	9.55
7	2023-02-28_NPB1	2023-02-28 00:00:00	NPB1	ostr~	0	NA	NA	NA
8	2023-05-16_NVBR	2023-05-16 00:00:00	NVBR	ostr~	0.133	9.29	NA	9.83
9	2022-02-23_NVBR	2022-02-23 00:00:00	NVBR	ostr~	0	8.86	45.2	10.2
10	2021-09-01_NPB2	2021-09-01 00:00:00	NPB2	ostr~	0	NA	NA	NA

i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
 # Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
 # Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
 # comments <chr>, site_full <chr>, log_ave_lit <dbl>

c. Code your figure

```
# base layer
ostracods <- ggplot(data = ostracods,
                    # x-axis: site
                    # y-axis: mean abundance/L
                    mapping = aes(x = site_full,
                                  y = log_ave_lit)) +
# first layer: points to represent observations
geom_jitter(height = 0,
            shape = 21,
            width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
            fun = "median",
            size = 4,
            color = "blue") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(10)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracods") +
# different theme
theme_bw()

# displaying object
ostracods
```



d. Create a multipanel figure

```
salinity / (copepods + ostracods)
```